Predicting the Outcome of Interstate Wars

A Bayesian and Logistic Regression Analysis Using Correlates of War Dataset, 1823-2010

By: Brian Kruchkow

Abstract

This paper develops two complementary quantitative models for predicting the outcome of interstate wars. The first is a **Bayesian Beta-Binomial model** that estimates a country's overall probability of winning a war based on its historical win-loss record. The second is a **logistic regression model** that estimates the probability of victory in any individual conflict based on the relative military and economic capabilities of the two belligerents at the time the war begins. Both models are applied to 270 directed combatant dyads drawn from 91 interstate wars recorded in the Correlates of War (COW) Inter-State War Dataset between 1823 and 2010, combined with capability data from the COW National Material Capabilities dataset. We find that a five-component capability model -- comprising military expenditure, military personnel, iron and steel production, primary energy consumption, and total population -- achieves an area under the ROC curve (AUC) of 0.790 on 151 decisive dyads, substantially outperforming the single-index CINC score (AUC 0.560).

1. Introduction

Can we predict who wins a war before it is fought? The question has interested strategists, historians, and political scientists for centuries. Classical military theory holds that material strength, comprising the size of armies, the output of factories, and the wealth of nations, can create a strong advantage for battlefield success. Modern quantitative IR research has largely confirmed this intuition: states with more resources win wars more often than states with fewer, though the relationship is far from deterministic.

This paper formalises that intuition into two explicit statistical models. We work with two widely-used datasets maintained by the **Correlates of War (COW) Project**: the *Inter-State War Dataset v4.0*, which catalogues every interstate war between 1816 and 2010 and records which participant won, and the *National Material Capabilities dataset v6.0*, which records six annual indicators of national power for every state in the international system from 1816 onwards.

The goal is not only to build mathematical models but also to explain how those models represent the real world and the conflicts that occur. This paper aims to make Bayesian statistics easily accessible, as the models rely on a Bayesian framework of prior and posterior distributions, and the innovation of a high-density function to model uncertainty for victory.

2. Data

2.1 The War Dataset

The COW Inter-State War Dataset records 95 wars fought between recognised nation-states. We apply two data cleaning steps before analysis. First, we **exclude the Seven Weeks War of 1866** entirely. That war involved the wholesale absorption of several micro-states (Hanover, Saxony, Hesse Electoral, Hesse Grand Ducal) whose NMC records are fragmentary and whose political existence was extinguished by the conflict, making any capability-based prediction exercise misleading. Second, we **recode the Vietnam War outcome for the United States as a Draw** (the dataset codes it as a US win; we follow the scholarly consensus that the 1975 outcome represents a strategic loss for US war aims, consistent with the armistice framing of the 1973 Paris Accords).

After these adjustments, the dataset contains **91 wars** and **270 directed combatant dyads**. A directed dyad is one row per ordered pair of belligerents: so the Crimean War, which involved the United Kingdom, France, Sardinia, and the Ottoman Empire all fighting Russia, contributes multiple rows: one for (UK, Russia), one for (France, Russia), and so on. The dependent variable is the outcome for the first-named state (State A): Win, Loss, or Draw.

2.2 The Capability Dataset

The National Material Capabilities (NMC) v6.0 dataset provides six annual indicators for each state:

Variable	Unit	Description
milex	Thousands USD (1914)	Military expenditure -- money spent on the armed forces in a given year
milper	Thousands of personnel	Military personnel -- total size of the active armed forces
irst	Thousands of metric tons	Iron and steel production -- a proxy for heavy industrial capacity
pec	Thousands of coal-ton equivalents	Primary energy consumption -- total energy use, a measure of industrial output
tpop	Thousands of people	Total population -- relevant to both economic output and manpower reserves
upop	Thousands of people	Urban population -- correlated with industrial development (not used in our models)

Table 1. NMC v6.0 variables used in this study.

Missing values in the NMC dataset are coded as `-9`. We replace these with *not available* and, when a capability reading is missing for the exact year a war began, we search up to two years forward and backward and use the nearest available observation. If no observation can be found within that window, the dyad is excluded from the logistic regression model (but not from the Bayesian model, which does not require capability data).

2.3 The Composite Index of National Capability (CINC)

The NMC dataset also provides a pre-computed composite score called the CINC. For any state in year t, the CINC is calculated as follows. First, compute the world total for each of the six indicators. Then, for each state, express its value as a share of that world total. Finally, average the six shares:

```
CINC_i = (1/6) x (milex_share + milper_share + irst_share +
pec_share + tpop_share + upop_share)
```

For example, if a country accounts for 5% of world military spending, 3% of world military personnel, 8% of world iron production, 6% of world energy, 4% of world population, and 7% of world urban population, its CINC score is (5+3+8+6+4+7)/6 = 5.5% = **0.055**. The CINC scores of all states in any given year sum to 1.0.

The COW project presents CINC as a convenient single-number summary of national power, and it is widely used in the quantitative IR literature. However, as we will show in Section 5, CINC has limited predictive power for war outcomes (AUC = 0.560), barely above the 0.5 you would get by flipping a coin. This is because **CINC is an unweighted mean of its own inputs, but not all inputs correlate to victory in a conflict**. We therefore build our capability model using individual variable ratios rather than CINC.

2.4 Excluding Urban Population Variable

The Urban Population variable was excluded from the National Material Capabilities 5-Component Model due to three reasons. First, the correlation between total population and urban population was 0.868. Since urban population is largely a function of total population, including both variables would create multicollinearity, making it difficult to isolate the individual effects of either total population or urban population on the dependent variable, victory in war.

Second, adding the urban population worsened the model's performance. Using a 5-component model excluding the urban population gave an area under the curve (AUC) of 0.786 on 151 dyads, while including the urban population variable gave an AUC of 0.769 on 145 dyads. The AUC drop of 0.017 is a sign of an unhelpful variable that adds noise rather than signal.

Lastly, because data on urban populations were missing in six of the dyads, the model would lose training cases. Those six cases represented 4% of the dyads, further worsening the model.

3. Model 1: Bayesian Estimation of Win Probability

The first question we can ask is simple: based on everything a country has done in past wars, what fraction of its wars does it win? The Bayesian model answers this question and tells us how confident we should be in that estimate.

3.1 Starting with Probability

Suppose we want to estimate the probability p that a particular country wins any war it fights. We could just count: if it has won 8 out of 12 wars, then our estimate is 8/12 = 0.667. But what if we have only seen the country fight 2 wars? A 2-0 record might be luck. The Bayesian approach handles this well by starting with a prior belief before any data are observed, and then updating that belief each time we see a new outcome.

3.2 The Beta Distribution as a Prior

We need a probability distribution over p: that is, not just a single number, but a description of how plausible each possible value of p is. The Beta distribution is a natural choice because it represents data coded with clear win/loss criteria.

The Beta distribution is controlled by two positive numbers, *alpha (α)* and *beta (β)*, which behave like imaginary prior observations. Think of *alpha* as your imaginary number of prior wins and *beta* as your imaginary number of prior losses.

$$p \sim \text{Beta}(\alpha, \beta)$$

The probability (p) follows a Beta distribution with two shape parameters, *alpha (α)* and *beta (β)*. Setting $\alpha = \beta = 1$ gives a flat, uniform prior: before seeing any data, you consider every value of

p from 0 to 1 equally plausible. There could also be bias toward winning outcomes, in which case alpha would be larger, or losing outcomes, in which case *beta* would be larger.

3.3 Updating the Prior: The Beta-Binomial Model

The Beta distribution is a conjugate prior for a binomial distribution with parameters (α, β), meaning the Beta distribution that we use to track wins and losses will have the same parameters as the output distribution that we are interested in to forecast future wins and losses. When we observe W wins and L losses, we can update our Beta prior by simply adding the new observations to alpha and beta:

```
Posterior distribution:  p | data ~ Beta(alpha + W,  beta + L)
```

The posterior distribution can be read as: the probability of a win, given the data, follows a Beta distribution with parameters *alpha* (prior wins) plus any observed wins, and *beta* (prior losses) plus any observed losses. This posterior distribution is still a Beta distribution, just with larger parameters. The posterior mean, which is our best single estimate of p, is:

```
E[p] = (alpha + W) / (alpha + beta + W + L)
```

The expected probability, E[p], is equal to alpha plus wins, divided by alpha plus beta, and the sum of all observations (W+L).

Intuition: Why use a prior?

Imagine you meet someone who has flipped a coin once and got heads. The naive estimate of their coin's bias toward heads is 100%. The Bayesian approach with a prior of α =1, β = 1 gives (1+1)/(1+1+1) = 2/3, which better reflects our background knowledge that most coins are roughly fair.

The same logic applies to wars. A country with 1 win and 0 losses is not necessarily unbeatable; the Bayesian prior pulls the estimate back toward uncertainty.

Worked example -- United Kingdom: The UK fought 21 directed dyads in our dataset. Using only decisive outcomes (excluding draws), it recorded 17 wins (W = 17) and 1 loss (L = 1). With a flat prior (alpha = beta = 1):

```
E[p] = (1 + 17) / (1 + 1 + 17 + 1) = 18 / 20 = 0.900
```

The Bayesian model estimates that the UK wins 90% of its wars. Notice how the prior (two imaginary observations) slightly pulls the estimate away from the raw 17/18 = 94.4%.

3.4 The 95% Highest Density Interval

A single number is useful, but we also want to know how certain we are. The 95% Highest Density Interval (HDI) is the narrowest interval that contains 95% of the posterior probability mass. Intuitively, it is the range of values of p that are most credible given the data.

Unlike a classical confidence interval, the HDI has a direct probabilistic interpretation: we believe with 95% probability that the true win probability lies somewhere in this range. We

compute the HDI numerically by finding the two values on the Beta distribution that together leave only 5% of the probability in the tails.

For the UK example above, the posterior is Beta(18, 2). The 95% HDI is approximately [**0.684, 0.997**], reflecting genuine uncertainty, despite the strong record, as 18 total observations are not a large sample.

3.5 Sequential Updating

One of the most useful properties of the Bayesian framework is that updates can be applied sequentially: we process wars one by one, and the posterior after each war becomes the prior for the next. This lets us trace how a country's estimated probability evolves through time.

The mathematics are identical at each step. If after processing k wars our posterior is Beta(alpha_k, beta_k), and the next war results in a win, then:

```
alpha_(k+1) = alpha_k + 1      (win)
beta_(k+1)  = beta_k           (no new loss)
```

If it results in a loss the update is symmetric: beta_(k+1) = beta_k + 1 and alpha_(k+1) = alpha_k.

3.6 Handling Draws

Our dataset records three possible outcomes: Win, Loss, and Draw. Draws are ambiguous, they are neither a win nor a loss. We offer two approaches, and the default is to exclude draws entirely (they contribute nothing to the win/loss count).

An alternative is to count each draw as half a win and half a loss, adding 0.5 to both alpha and beta. This preserves the total count of wars while acknowledging that a draw is somewhere between victory and defeat. Formally:

```
If draw:   alpha = alpha + 0.5,   beta = beta + 0.5
```

The two approaches rarely produce dramatically different estimates unless a country has a large proportion of draws (as China does, where most dyads ended in draws).

4. Model 2: Logistic Regression on Capability Ratios

The Bayesian model tells us how a country *has performed* historically. The second model asks a different question: given what we know about the *relative capabilities* of two states at the moment they go to war, how likely is the stronger state to win *this particular conflict*?

4.1 The Key Idea: Capability Ratios

Rather than using raw capability values (which differ enormously across centuries), we convert every variable into a ratio. For any pair of states A and B, the ratio for variable x is:

```
Ratio x = x^A / (x^A + x^B)
```

This ratio always lies between 0 and 1. A ratio of **0.5** means both states have equal capability on that dimension. A ratio of **0.8** means State A controls 80% of the combined capability. A ratio of **0.2** means State A has only one-fifth as much as State B.

Edge case: When both states have a value of 0 for some indicator (for example, neither has any recorded iron production), we assign a ratio of 0.5, as neither has an advantage.

4.2 The Logistic Regression Model

We want to predict a binary outcome, Win (1) or Loss (0), from a set of numeric input variables. Logistic regression is the standard tool. It works in two steps.

Step 1 -- Compute the log-odds (the "logit"): We take a weighted sum of the input variables plus a constant intercept B0:

```
z = B0 + B1 x milex_ratio + B2 x milper_ratio + B3 x irst_ratio +
B4 x pec_ratio + B5 x tpop_ratio
```

Step 2 -- Convert to a probability via the sigmoid function: The sigmoid transforms any real number z into the range (0, 1):

```
P(win) = sigma(z) = 1 / (1 + e^(-z))
```

Why the sigmoid?

We cannot use plain linear regression to predict a probability because a linear equation can produce values below 0 or above 1, which are impossible for a probability.

The sigmoid function maps any value on the number line to the interval (0, 1):

```
z = -inf  -->  P = 0.00    (guaranteed loss)
z =  0    -->  P = 0.50    (even odds)
z = +inf  -->  P = 1.00    (guaranteed win)
```

4.3 Standardising the Input Variables

The five capability ratios do not all have the same scale of variation. To make the regression coefficients comparable in magnitude, so we can say which variables are more "influential", we standardise each ratio before fitting the model:

```
x* = (x - mean(x)) / std(x)
```

Here mean(x) and std(x) are computed from the training data. After standardisation, a coefficient of +1.0 means "a one standard deviation increase in this ratio is associated with a z-score increase of 1.0", making coefficients directly comparable across variables.

4.4 Training the Model

The coefficients B0 through B5 are estimated by **Maximum Likelihood Estimation (MLE)** where we choose the values that make the observed outcomes (Win or Loss) as probable as possible under the model. This is standard logistic regression, implemented using scikit-learn's L2-regularised solver.

We train on **151 directed dyads**, those where both states had complete NMC records for all five variables in the year the war began, and where the outcome was decisive (Win or Loss, not Draw). Each dyad appears once, deduplicated on the directed-dyad index. For multi-year wars we take the first observation.

4.5 Fitted Coefficients

Variable	Coefficient	Train Mean	Interpretation
Intercept (B0)	+0.227	--	Baseline log-odds when all ratios are at their mean
Military expenditure (B1)	+0.314	0.449	More spending -> higher win probability
Military personnel (B2)	-0.834	0.464	More troops -> lower win probability (see Section 4.6)
Iron/steel production (B3)	+0.859	0.468	Largest positive predictor -- industrial capacity matters most
Energy consumption (B4)	+0.720	0.465	Second strongest positive predictor
Total population (B5)	-0.683	0.442	Larger population -> lower win probability (see Section 4.6)

Table 2. Fitted logistic regression coefficients on standardised capability ratios.

4.6 Why Do Milper and Population Have Negative Coefficients?

At first glance, the negative signs on military personnel and population seem backwards, surely bigger armies and larger populations help? But this is not the case, as larger populations often try to compensate for lower economic output with increased manpower, which is not decisive in war.

Consider two states with equal iron production and energy consumption. The one with a *larger* army relative to its industrial base is, in effect, **maintaining more soldiers than its industry**

can effectively equip. In the historical record, states that had large conscript armies but modest industrial capacity, think Turkey in WWI, Russia in 1904, or China throughout the 19th century, tended to lose to smaller but industrially superior opponents.

The model is not saying "fewer soldiers is better." It is saying that, conditional on industrial output, a high milper ratio signals a capability mismatch that historically predicts defeat.

4.7 Worked Example: UK vs. Russia, Crimean War 1854

Let us trace through the full calculation for the United Kingdom (State A) against Russia (State B) in 1854.

Variable	UK value	Russia value	Raw ratio (UK / total)
milex (thou. USD)	76,329	31,358	76329/(76329+31358) = 0.709
milper (thou.)	234	1,100	234/(234+1100) = 0.175
irst (thou. MT)	3,119	231	3119/(3119+231) = 0.931
pec (thou. coal-eq)	70,914	141	70914/(70914+141) = 0.998
tpop (thou.)	27,658	70,600	27658/(27658+70600) = 0.282

Next we standardise each raw ratio. The standardisation parameters (mean and standard deviation) come from the training data:

```
milex*  = (0.709 - 0.449) / 0.347 = +0.749
milper* = (0.175 - 0.464) / 0.319 = -0.906
irst*   = (0.931 - 0.468) / 0.396 = +1.169
pec*    = (0.998 - 0.465) / 0.373 = +1.428
tpop*   = (0.282 - 0.442) / 0.296 = -0.541
```

Now compute the log-odds z by multiplying each standardised ratio by its coefficient and summing:

```
z = 0.227                                [intercept]
  + 0.314 x (+0.749)    = +0.235        [milex]
  + (-0.834) x (-0.906) = +0.756        [milper]
  + 0.859 x (+1.169)    = +1.004        [irst]
  + 0.720 x (+1.428)    = +1.028        [pec]
  + (-0.683) x (-0.541) = +0.370        [tpop]
  = 3.620
```

Finally, convert to a probability:

```
P(UK wins) = 1 / (1 + e^(-3.620))
           = 1 / (1 + 0.0267)
           = 1 / 1.0267
           = 0.974  -->  97.4%
```

The model assigns the UK a **97.4% probability of winning** the Crimean War against Russia in 1854. The UK did, in fact, win. The dominant driver is the UK's overwhelming advantage in energy consumption (pec ratio = 0.998) as it was the world's leading industrial power at the time, and in iron and steel output.

5. Model Evaluation

5.1 The AUC Metric

We evaluate the logistic model using the Area Under the Receiver Operating Characteristic Curve (AUC). Intuitively, the AUC is the probability that the model assigns a higher probability of winning to a dyad that actually won than to a dyad that actually lost, when one is selected at random from each group.

An AUC of **0.5** means the model does no better than random. An AUC of **1.0** means the model is perfectly discriminating. An AUC of **0.7-0.8** is generally considered "acceptable" in social science applications; above 0.8 is considered "excellent".

5.2 Results

Model	AUC	n (dyads)	Notes
Random baseline	0.500	--	Coin flip -- no model at all
CINC ratio only	0.560	216	Single composite score; barely above random
5-component model (milex, milper, irst, pec, tpop)	0.790	151	Our preferred model

Table 3. AUC scores for each model variant.

The 5-component model achieves AUC = **0.790**, well above both the random baseline and the CINC-only alternative. The inferior performance of the CINC model reflects the problem discussed in Section 2.3: CINC's equal weighting of all six components obscures the differential predictive power of industrial variables versus population and manpower, and CINC double weights population with the urban and total population, introducing multicollinearity.

Variable	AUC	Interpretation
Iron and steel production (irst)	0.673	Strongest single predictor
Primary energy consumption (pec)	0.680	Essentially tied with irst
Military expenditure (milex)	0.583	Moderate signal
Military personnel (milper)	0.523	Weak -- close to random
Total population (tpop)	0.515	Nearly random when used alone
CINC (composite)	0.560	Average of six components -- mediocre

Table 4. AUC for each individual capability ratio.

As a diagnostic, we fit a separate logistic regression for each variable individually. The results confirm that iron/steel production and energy consumption carry most of the predictive signal.

5.3 The ROC Curve and Area Under the Curve (AUC)

The logistic regression model assigns each directed dyad a predicted probability of winning, a number between 0 and 1. To turn that probability into a binary prediction (*win* or *not win*), we must choose a **classification threshold**: we predict a win whenever the predicted probability exceeds the threshold, and a loss otherwise. Different threshold choices yield different error rates, and the ROC curve summarizes them simultaneously.

For any fixed threshold, there are four possible outcomes for each dyad:

	Model predicts: Win	Model predicts: Loss
Actually won	True Positive (TP)	False Negative (FN)
Actually lost	False Positive (FP)	True Negative (TN)

The ROC curve is constructed by sweeping the threshold from 1 down to 0. At each step we compute two quantities:

True Positive Rate (TPR): also called Recall, it is the fraction of actual wins the model correctly identifies at that threshold.

```
TPR = TP / (TP + FN)
```

False Positive Rate (FPR): the fraction of actual losses the model incorrectly labels as wins.

```
FPR = FP / (FP + TN)
```

The ROC curve plots TPR on the vertical axis against FPR on the horizontal axis for every possible threshold. A model that predicts randomly traces the diagonal from (0,0) to (1,1). A perfect model would jump immediately to the top-left corner (TPR=1, FPR=0) and stay there. The **Area Under the ROC Curve (AUC)** is the literal area enclosed by the curve and the horizontal axis. It has a clean probabilistic interpretation: it is the probability that, if we pick one dyad that the stronger state actually won and one that it actually lost at random, the model assigns a higher probability of winning to the winner. An AUC of 0.5 means the model has no discriminating power; an AUC of 1.0 means it is perfectly discriminating.

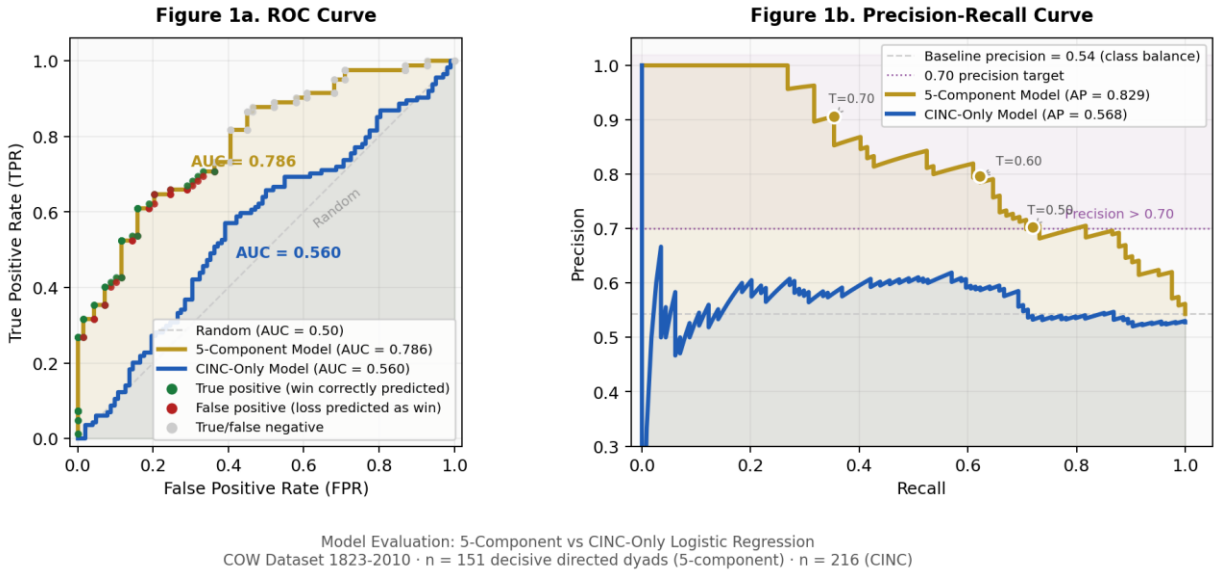

Figure 1. Left panel: ROC curves for the 5-Component logistic model (gold, AUC = 0.786) and the CINC-only model (blue, AUC = 0.560). The diagonal dashed line represents a random classifier (AUC = 0.50). Colored dots show individual war dyads at threshold T = 0.50: green = true positives, red = false positives. Right panel: Precision-Recall curves for the same two models. The purple shaded region marks the zone where precision exceeds 0.70. Labelled markers (T=0.50, T=0.60, T=0.70) show how precision and recall trade off as the classification threshold rises. Source: COW Inter-State War Dataset v4.0; NMC v6.0. n = 151 decisive directed dyads (5-component); n = 216 (CINC).

Figure 1a shows the ROC curves for both models. The five-component model (AUC = 0.786) bows considerably further toward the top-left corner than the CINC-only model (AUC = 0.560). The CINC curve barely clears the random diagonal, confirming that the composite index adds little discriminating power beyond what chance would provide. The five-component model falls in the range conventionally described as *"acceptable"* to *"good"* in social science applications, where an AUC above 0.70 is considered meaningful and above 0.80 is considered excellent.

5.4 Precision

While the AUC summarises the model across all thresholds simultaneously, practical forecasting requires choosing a single threshold. The two most important per-threshold statistics are **Precision** and **Recall**.

Precision answers the question: *of all the wars the model predicted the stronger state would win, what fraction did it actually win?*

```
Precision = TP / (TP + FP)
```

Precision measures the reliability of a positive prediction. A precision of 0.80 means that eight out of ten wars the model labels as predicted wins are genuine wins, the other two are false alarms, cases where the more capable state nonetheless lost. Precision is also called **Positive Predictive Value**.

In a forecasting context, precision governs the **cost of acting on a wrong prediction**. If a government or analyst uses the model to allocate resources, commit allies, or issue deterrence signals ahead of a conflict, a false positive, treating a likely loss as a likely win, could lead to catastrophic miscalculation. Higher precision means fewer such errors, at the price of making fewer predictions overall.

5.5 Recall

Recall answers the complementary question: *of all the wars the stronger state actually won, what fraction did the model correctly identify in advance?*

```
Recall = TP / (TP + FN)
```

Recall measures **coverage**, how much of the truth the model catches. A recall of 0.72 at threshold 0.50 means the model would have correctly anticipated roughly seven out of every ten eventual wins in the historical record. The remaining 23 per cent are **false negatives**: wars the stronger state won that the model did correctly identify.

False negatives carry a different kind of cost from false positives. Missing a predicted win means failing to anticipate the outcome of a conflict that the material balance of forces would have correctly foreshadowed. In a policy context, this could mean underestimating one's own position or failing to advise a state that its military prospects are actually strong.

5.6 The Precision-Recall Trade-off and the F1 Score

Precision and Recall are in fundamental tension. Raising the classification threshold, requiring the model to be more confident before predicting a win, increases Precision (fewer false alarms) but decreases Recall (more missed wins). This trade-off is visualized directly in Figure 1b, the **Precision-Recall curve**. Each point on the curve corresponds to a different threshold; moving right along the curve represents a lower threshold (more predictions made, lower precision, higher recall).

The F1 score combines Precision and Recall into a single number by taking their harmonic mean:

```
F1 = 2 x (Precision x Recall) / (Precision + Recall)
```

The harmonic mean penalizes extreme imbalances: a model with perfect Precision but zero Recall (or vice versa) scores F1 = 0, not 0.50. F1 = 1.0 is the theoretical maximum. For the five-component model, the F1 score peaks near threshold 0.50 and declines as the threshold rises, because the gains in Precision are increasingly outweighed by the losses in Recall.

Table 5 reports Precision, Recall, and F1 for a range of classification thresholds for the five-component model.

Threshold	Precision	Recall	F1	TP	FP	FN
0.40	0.636	0.915	0.750	75	43	7
0.50	0.702	0.720	0.711	59	25	23
0.55	0.750	0.659	0.701	54	18	28
0.60	0.797	0.622	0.699	51	13	31
0.65	0.826	0.463	0.594	38	8	44
0.70	0.906	0.354	0.509	29	3	53
0.75	1.000	0.232	0.376	19	0	63

Table 5. Classification performance of the 5-component logistic model at varying probability thresholds. TP = true positives; FP = false positives; FN = false negatives. All metrics computed on the training set (n = 151 decisive directed dyads). The purple shaded region in Figure 1b marks thresholds where precision exceeds 0.70.

Three threshold regimes are worth distinguishing. At **T = 0.40**, the model casts the widest net: it makes 118 predictions and correctly calls 75 wins (recall 0.915), but accepts 43 false positives and a precision of only 0.636, which is well below what most scholars would regard as acceptable. At **T = 0.50**, the conventional default, precision reaches 0.702 and recall 0.720, yielding an F1 score of 0.711. This is the best single-threshold balance the model achieves. At **T = 0.70**, precision climbs to 0.906 (fewer than one false positive in ten), but recall collapses to 0.354, which means the model is now so conservative that it misses nearly two-thirds of the actual wins in the dataset.

5.5 Implications for Forecasting Future Conflicts

These metrics have concrete implications for how the model should be used, and how cautiously its predictions should be interpreted, when applied to a conflict that has not yet occurred.

What the model can and cannot claim

At the conventional threshold of 0.50, the five-component model correctly identifies approximately 72 per cent of historical decisive conflicts whose stronger side prevailed. Put differently, if the model had been available to analysts at the outset of each war in the dataset and applied prospectively, it would have correctly forecast the direction of 59 of the 82 eventual wins among decisive dyads, while generating 25 false alarms. This is a substantially better record than the CINC model, which adds little to chance, and better than the baseline of predicting that the materially stronger state always wins (which would achieve precision equal only to the class balance, approximately 0.54).

At the same time, the model operates entirely on **pre-war material capabilities**. It does not yet observe alliances, terrain, leadership quality, the element of surprise, domestic political will, or the trajectory of the conflict once fighting begins. The false negatives in Table 5, wars the materially stronger state won but the model failed to predict, are not random noise; many reflect cases where an industrially dominant power was fighting at a geographic or political disadvantage, or where a smaller but more motivated force extracted unexpected performance from its equipment. The model does not yet have the data to model these factors.

Threshold choice as an analytic decision

When applying the model prospectively, the appropriate threshold depends on the **asymmetry of error costs** in the specific context. Three scenarios illustrate the choice:

Academic inference (T = 0.50). A researcher using the model to explain patterns across many conflicts is equally concerned with false positives and false negatives. The default threshold of 0.50 maximizes F1 and provides the best overall balance. The model is best understood as identifying the direction the balance of capabilities points, not as a deterministic predictor of individual outcomes.

Policy advising with high false-positive costs (T = 0.60 to 0.70). An analyst advising a government that is considering intervention alongside a weaker ally, and where a false positive could lead to disastrous commitment, should raise the threshold. At T = 0.60, precision reaches 0.797 and at T = 0.70 it reaches 0.906: the model only predicts a win when the capability advantage is overwhelming. The cost is that genuine wins are missed (recall falls to 0.622 and 0.354 respectively), but the predictions that are made are highly reliable.

6. Discussion

6.1 What the Two Models Tell Us

The Bayesian and logistic models answer fundamentally different questions which complement each other:

- Bayesian model: "Given this country's history, how often does it win wars in general?"
- Logistic model: "Given the balance of capabilities at the start of this specific war, how likely is this country to win?"

The two models can disagree. A historically strong country (high Bayesian posterior) might enter a particular war at a capability disadvantage (low logistic probability). Conversely, an industrial giant with a modest win record might be heavily favoured in a specific matchup.

6.2 The Industrial Revolution Signal

The dominance of iron/steel production and energy consumption in the logistic model is consistent with a well-established finding in military history: from the Crimean War onwards, industrial capacity was the primary determinant of battlefield outcome. The ability to produce artillery, ammunition, railroads, and ships at scale, which are all proxied by irst and pec, was more decisive than the size of the army one could field.

This helps explain several famous results in our dataset. France lost to Prussia in 1870 despite a larger army, because Germany's industrial base (captured in its superior irst and pec ratios)

gave it decisive logistical advantages. Japan defeated Russia in 1904-05 partly because its newer industrial infrastructure offset Russia's larger population.

6.3 Limitations

- The model only uses pre-war capability data. It does not capture alliances, terrain, leadership, surprise, or the course of the war itself.
- The dataset covers only interstate wars; civil wars, colonial wars, and insurgencies are excluded.
- Wars involving many belligerents are decomposed into directed dyads, which treat each pair independently and ignore coalition dynamics.
- NMC data quality is uneven, especially for 19th-century conflicts in Asia, Africa, and Latin America.
- The model is trained and evaluated on the same historical dataset. With only 151 complete observations, out-of-sample validation would be valuable but requires a separate test set.

7. Conclusion

We have built two interpretable quantitative models for predicting war outcomes. The Bayesian Beta-Binomial model provides a posterior estimate of a country's baseline win probability that updates as new wars are observed, with uncertainty quantified through a highest density interval. The logistic regression model predicts the outcome of individual conflicts from five capability ratios, achieving AUC = 0.790 on 151 decisive directed dyads.

Both models are deliberately transparent for academic purposes. The complete training dataset is provided in the appendix so that readers can reproduce, extend, or critique results.

The central finding is that industrial output, measured through iron and steel production and primary energy consumption, is a stronger predictor of war outcomes than army size or even the composite CINC score. This result holds across nearly two centuries of interstate conflict and is consistent with the broader literature on industrial mobilisation and military effectiveness.

Appendix A. Training Dataset

This appendix contains the 151 directed dyads used to train the 5-component logistic regression model. *State A* is the state whose win probability is being estimated. All ratio columns express State A's share of the combined value: `ratio = A / (A + B)`. A ratio above 0.5 means State A has the larger value. The *Outcome* column is the dependent variable (Win = 1, Loss = 0). Draws are excluded. Source: COW Inter-State War Dataset v4.0 and NMC v6.0.

War	Year	State A	State B	Outcome	Milex	Milper	IRST	PEC	TotPop
Mexican-American War	1846	Mexico	United States of America	Loss	0.3572	0.339	0.0	0.0025	0.2625
Austro-Sardinian War	1848	Italy	Austria-Hungary	Loss	0.3603	0.1461	0.1124	0.0588	0.1192
Crimean War	1854	United Kingdom	Russia	Win	0.7088	0.1754	0.931	0.998	0.2815
Crimean War	1854	France	Russia	Win	0.4919	0.2857	0.7695	0.9871	0.3391
Crimean War	1855	Italy	Russia	Win	0.0533	0.0331	0.0738	0.3858	0.0657
Italian Unification War	1859	Italy	Austria-Hungary	Win	0.2003	0.1299	0.0592	0.0815	0.1185
Italian Unification War	1859	Austria-Hungary	France	Loss	0.414	0.4329	0.269	0.1425	0.5073
Second Schleswig-Holstein War	1864	Denmark	Austria-Hungary	Loss	0.0603	0.097	0.0	0.0735	0.0449
Second Schleswig-Holstein War	1864	Denmark	Germany	Loss	0.1091	0.1311	0.0	0.0114	0.0818
Naval War	1866	Spain	Peru	Loss	0.8262	0.856	1.0	1.0	0.8619
Franco-Prussian War	1870	France	Germany	Loss	0.5787	0.5863	0.483	0.3967	0.552
Conquest of Egypt	1882	United Kingdom	Egypt	Win	0.9957	0.9427	1.0	1.0	0.8613
Greco-Turkish War (1897)	1897	Turkey	Greece	Win	0.8663	0.9458	1.0	0.5639	0.9185
Spanish-American War	1898	Spain	United States of America	Loss	0.0901	0.3933	0.0214	0.0186	0.2002
Boxer Rebellion	1900	China	France	Loss	0.2318	0.6169	0.0	0.0048	0.9162
Boxer Rebellion	1900	China	United States of America	Loss	0.2279	0.8881	0.0	0.0009	0.8483
Boxer Rebellion	1900	China	United Kingdom	Loss	0.0929	0.6725	0.0	0.0013	0.9118
Boxer Rebellion	1900	Russia	China	Win	0.7788	0.5331	1.0	0.9932	0.238
Boxer Rebellion	1900	China	Japan	Loss	0.4791	0.846	0.0	0.0283	0.9066
Sino-Russian War	1900	Russia	China	Win	0.7788	0.5331	1.0	0.9932	0.238
Russo-Japanese War	1904	Russia	Japan	Loss	0.4278	0.8418	0.9788	0.7579	0.7543
Third Central American War	1906	Guatemala	Honduras	Win	0.8071	0.875	0.5	0.5	0.6688
Third Central American War	1906	Guatemala	El Salvador	Win	0.6929	0.7	0.5	0.5	0.5309
Italian-Turkish War	1911	Italy	Turkey	Win	0.7254	0.4641	1.0	0.9337	0.5756
First Balkan War	1912	Yugoslavia	Turkey	Win	0.0911	0.114	0.5	0.1674	0.1061
First Balkan War	1912	Greece	Turkey	Win	0.0923	0.1007	0.5	0.3575	0.0959
First Balkan War	1912	Bulgaria	Turkey	Win	0.1206	0.202	0.5	0.2465	0.147
Second Balkan War	1913	Yugoslavia	Bulgaria	Win	0.4243	0.0513	0.5	0.4276	0.4002
Second Balkan War	1913	Bulgaria	Romania	Loss	0.3336	0.8445	0.5	0.1353	0.3797
Second Balkan War	1913	Greece	Bulgaria	Win	0.7523	0.2525	0.5	0.6554	0.3788
Second Balkan War	1913	Bulgaria	Turkey	Loss	0.0606	0.7107	0.5	0.2396	0.1742

World War I	1914	Austria-Hungary	Yugoslavia	Loss	0.996	0.9534	1.0	0.992	0.9289
World War I	1914	Belgium	Germany	Win	0.0055	0.1177	0.0918	0.0818	0.1018
World War I	1914	Austria-Hungary	Russia	Win	0.5487	0.3884	0.3262	0.401	0.231
World War I	1914	United Kingdom	Germany	Win	0.4847	0.3816	0.366	0.5332	0.4045
World War I	1914	Germany	Russia	Win	0.6756	0.3949	0.7556	0.7946	0.2791
World War I	1914	France	Germany	Win	0.4089	0.4779	0.1687	0.194	0.3289
World War I	1915	Italy	Austria-Hungary	Win	0.05	0.4539	0.2745	0.2474	0.407
World War I	1915	Bulgaria	Yugoslavia	Loss	0.5501	0.3684	0.5	0.3275	0.5479
World War I	1915	United Kingdom	Bulgaria	Win	0.9974	0.9889	1.0	0.9992	0.9005
World War I	1915	Yugoslavia	Germany	Win	0.0019	0.0215	0.0	0.002	0.0562
World War I	1915	Bulgaria	France	Loss	0.0034	0.0097	0.0	0.0043	0.128
World War I	1915	Austria-Hungary	France	Loss	0.3635	0.432	0.7059	0.4157	0.6137
World War I	1915	Austria-Hungary	United Kingdom	Loss	0.3021	0.4663	0.2349	0.1145	0.5446
World War I	1915	Turkey	France	Loss	0.0128	0.1365	0.0	0.0124	0.358
World War I	1916	Bulgaria	Italy	Loss	0.0749	0.2021	0.0	0.021	0.1199
World War I	1916	Italy	Germany	Win	0.032	0.4082	0.0786	0.0461	0.3515
World War I	1916	Austria-Hungary	Romania	Win	0.9937	0.799	1.0	0.8624	0.858
World War I	1916	Romania	Germany	Win	0.0031	0.044	0.0	0.0106	0.1155
World War I	1916	Bulgaria	Russia	Win	0.0031	0.0684	0.0	0.004	0.0268
World War I	1916	Romania	Bulgaria	Win	0.5389	0.2087	0.5	0.9115	0.6389
World War I	1917	Greece	Austria-Hungary	Win	0.0119	0.0477	0.0	0.0558	0.0794
World War I	1917	Greece	Germany	Win	0.0048	0.0077	0.0	0.002	0.0645
World War I	1917	Bulgaria	Greece	Loss	0.2793	0.475	0.5	0.3952	0.5208
World War I	1917	United States of America	Germany	Win	0.0843	0.1069	0.7471	0.7819	0.6052
World War I	1917	Germany	Portugal	Loss	0.9989	0.9773	1.0	0.9968	0.9181
Estonian Liberation War	1919	Finland	Russia	Win	0.0054	0.0239	0.0	0.0069	0.0215
Franco-Turkish War	1919	France	Turkey	Win	0.9498	0.9483	1.0	0.9875	0.6273
Hungarian Adversaries War	1919	Hungary	Romania	Loss	0.2806	0.0846	0.0	0.5481	0.4477
Hungarian Adversaries War	1919	Hungary	Czechoslovakia	Loss	0.306	0.1525	0.0	0.1279	0.3703
Latvian Liberation War	1919	Estonia	Russia	Win	0.0028	0.0102	0.0	0.0965	0.0077
Latvian Liberation War	1919	Russia	France	Loss	0.6907	0.396	0.1334	0.2487	0.8165
Second Greco-Turkish War	1919	Turkey	Greece	Win	0.3864	0.6898	0.5	0.6226	0.8025
Latvian Liberation War	1920	Poland	Russia	Win	0.0113	0.2398	0.8349	0.7329	0.1607
Lithuanian-Polish War	1920	Lithuania	Poland	Loss	0.2892	0.0494	0.0	0.0721	0.0721
Sino-Soviet War	1929	Russia	China	Win	0.9617	0.2485	0.9959	0.6679	0.2419
Manchurian War	1931	China	Japan	Loss	0.2483	0.8534	0.0079	0.4252	0.8815
Chaco War	1932	Paraguay	Bolivia	Loss	0.384	0.3	0.5	0.0	0.273
Second Sino-Japanese War	1937	Japan	China	Win	0.6979	0.1642	0.9932	0.6076	0.1211
Changkufeng War	1938	Russia	Japan	Win	0.7617	0.8089	0.7361	0.7482	0.7015
Nomohan War	1939	Japan	Russia	Loss	0.2212	0.3485	0.276	0.2495	0.2953
Russo-Finnish War	1939	Finland	Russia	Loss	0.0113	0.0203	0.0044	0.0119	0.0213
World War II	1939	Russia	Poland	Win	0.8607	0.8668	0.9721	0.8835	0.8276
World War II	1939	Germany	Poland	Win	0.9253	0.9091	0.9792	0.9144	0.6922
World War II	1940	Italy	Greece	Win	0.9071	0.9396	0.9934	0.9567	0.8587
World War II	1940	New Zealand	Italy	Win	0.016	0.0006	0.0	0.1072	0.0354

World War II	1940	Norway	Germany	Loss	0.0042	0.0012	0.0032	0.0139	0.0318
World War II	1940	France	Germany	Loss	0.2121	0.6024	0.17	0.1814	0.3125
World War II	1940	Netherlands	Germany	Loss	0.0079	0.019	0.0071	0.047	0.0896
World War II	1940	Australia	Italy	Win	0.2283	0.0478	0.37	0.491	0.1367
World War II	1940	Italy	France	Loss	0.0961	0.2463	0.3385	0.2337	0.5203
World War II	1940	United Kingdom	Germany	Win	0.3194	0.2361	0.3797	0.4753	0.3484
World War II	1940	Italy	United Kingdom	Loss	0.0575	0.6157	0.1462	0.0694	0.4797
World War II	1940	Belgium	Germany	Loss	0.0135	0.0242	0.0808	0.0919	0.0842
World War II	1940	Japan	France	Win	0.2461	0.1607	0.6084	0.5376	0.637
World War II	1940	United Kingdom	France	Win	0.6354	0.1694	0.7492	0.8034	0.5405
World War II	1941	France	Australia	Loss	0.5238	0.4825	0.7171	0.7206	0.5295
World War II	1941	Germany	Russia	Loss	0.8076	0.6279	0.5721	0.6424	0.3154
World War II	1941	Spain	Russia	Loss	0.0129	0.1104	0.0423	0.062	0.1167
World War II	1941	United Kingdom	Japan	Win	0.7938	0.4279	0.6464	0.7734	0.4004
World War II	1941	Japan	Canada	Loss	0.7383	0.9165	0.7356	0.5665	0.8626
World War II	1941	Japan	South Africa	Loss	0.9629	0.9624	0.9528	0.7491	0.8727
World War II	1941	Greece	Germany	Loss	0.0028	0.0551	0.0007	0.0026	0.0755
World War II	1941	Japan	United States of America	Loss	0.3174	0.6306	0.0835	0.0609	0.3512
World War II	1941	South Africa	Germany	Win	0.0039	0.0166	0.016	0.0781	0.1042
World War II	1941	Australia	Germany	Win	0.0187	0.0533	0.0754	0.0659	0.0728
World War II	1941	Germany	New Zealand	Loss	0.9958	0.9861	1.0	0.9918	0.9823
World War II	1941	Germany	Yugoslavia	Win	0.9938	0.9793	0.99	0.9882	0.8521
World War II	1942	New Zealand	Japan	Win	0.0484	0.0477	0.0	0.0332	0.022
World War II	1943	Italy	United States of America	Loss	0.0057	0.1584	0.021	0.0029	0.2469
World War II	1943	United States of America	Germany	Win	0.6168	0.5652	0.7952	0.8102	0.601
World War II	1943	Canada	Italy	Win	0.9029	0.303	0.6121	0.9453	0.2083
World War II	1943	Germany	Canada	Loss	0.921	0.9039	0.884	0.8238	0.885
World War II	1943	Italy	Germany	Win	0.0091	0.1966	0.0768	0.0122	0.3306
World War II	1944	Germany	France	Loss	0.9766	0.9137	0.8556	0.8854	0.6988
World War II	1945	Japan	Russia	Loss	0.3179	0.3278	0.1452	0.1847	0.2892
First Kashmir War	1947	India	Pakistan	Win	1.0	0.5307	1.0	0.9998	0.831
Israeli War of Independence	1948	Israel	Iraq	Win	0.7166	0.7328	0.5	0.1322	0.1503
Israeli War of Independence	1948	Israel	Lebanon	Win	0.8978	0.9659	0.5	0.7442	0.4195
Israeli War of Independence	1948	Israel	Syria	Win	0.7822	0.914	0.5	0.8205	0.2215
Israeli War of Independence	1948	Israel	Egypt	Win	0.5554	0.5705	0.5	0.0106	0.0428
Israeli War of Independence	1948	Israel	Jordan	Win	0.8655	0.914	0.5	1.0	0.4296
First Taiwan Straits Crisis	1954	China	Taiwan	Win	0.9743	0.8369	0.9889	0.9617	0.9856
Sinai War	1956	Israel	Egypt	Win	0.1814	0.4464	0.6667	0.2816	0.0719
Sinai War	1956	France	Egypt	Loss	0.9387	0.902	0.9996	0.9663	0.6507
Sinai War	1956	United Kingdom	Egypt	Loss	0.95	0.8911	0.9997	0.9837	0.686
Soviet Invasion of Hungary	1956	Russia	Hungary	Win	0.9921	0.9595	0.9718	0.9721	0.9527
Sino-Indian War	1962	India	China	Loss	0.0889	0.2581	0.4357	0.2437	0.3986
Second Kashmir War	1965	Pakistan	India	Win	0.1919	0.185	0.002	0.1206	0.1898
Vietnam War	1965	South Korea	Vietnam	Loss	0.2299	0.7023	0.6491	0.8575	0.6017
Vietnam War	1965	Republic of Vietnam	Vietnam	Win	0.5683	0.6882	0.0	0.0307	0.459

Vietnam War	1965	Australia	Vietnam	Win	0.6507	0.1715	0.9812	0.9458	0.3748
Vietnam War	1966	Vietnam	Philippines	Loss	0.7831	0.8505	1.0	0.2909	0.3714
Six Day War	1967	Israel	Jordan	Win	0.831	0.5556	1.0	0.9974	0.5712
Six Day War	1967	Egypt	Israel	Loss	0.5371	0.7458	0.7073	0.5846	0.9192
Six Day War	1967	Iraq	Israel	Loss	0.3362	0.5455	0.0	0.6045	0.7596
Six Day War	1967	Israel	Syria	Win	0.8369	0.4839	1.0	0.9886	0.3234
Vietnam War	1967	Thailand	Vietnam	Loss	0.2162	0.2654	0.0148	0.5578	0.6244
War of Attrition (Laos-Vietnam)	1967	Vietnam	Laos	Win	0.9221	0.8708	1.0	1.0	0.878
Football War	1969	El Salvador	Honduras	Win	0.5961	0.4	1.0	0.4326	0.5693
Yom Kippur War	1973	Syria	Israel	Loss	0.1107	0.4694	0.0	0.2293	0.6782
Yom Kippur War	1973	Israel	Egypt	Win	0.4942	0.25	0.2899	0.4795	0.0848
Turkish Invasion of Cyprus	1974	Greece	Turkey	Loss	0.4802	0.2447	0.3203	0.4736	0.1867
Turkish Invasion of Cyprus	1974	Turkey	Cyprus	Win	0.9841	0.9812	1.0	0.972	0.9838
Ogaden War	1977	Ethiopia	Somalia	Win	0.8098	0.8094	0.5	0.8659	0.8589
Ogaden War	1978	Cuba	Somalia	Win	0.9393	0.7955	1.0	0.9	0.6093
Ugandan-Tanzanian War	1978	Tanzania	Uganda	Loss	0.6294	0.913	0.0	0.7442	0.5771
Ugandan-Tanzanian War	1979	Tanzania	Libya	Win	0.0802	0.5096	0.5	0.3697	0.8723
Falklands War	1982	Argentina	United Kingdom	Loss	0.1465	0.3431	0.1672	0.1688	0.3396
Chad-Libya War	1987	Libya	Chad	Loss	0.9747	0.7521	0.5	0.988	0.4125
Gulf War (1991)	1991	Iraq	Italy	Loss	0.3955	0.5011	0.0	0.0993	0.246
Gulf War (1991)	1991	Saudi Arabia	Iraq	Win	0.7335	0.2868	1.0	0.9076	0.4719
Gulf War (1991)	1991	Canada	Iraq	Win	0.4696	0.1533	1.0	0.9191	0.6022
Gulf War (1991)	1991	Syria	Iraq	Win	0.2586	0.4621	1.0	0.4663	0.4036
Gulf War (1991)	1991	Egypt	Iraq	Win	0.1134	0.4774	1.0	0.6976	0.7465
Gulf War (1991)	1991	Iraq	France	Loss	0.2568	0.4616	0.0	0.0916	0.245
Gulf War (1991)	1991	Iraq	United States of America	Loss	0.0469	0.1838	0.0	0.0088	0.0683
Gulf War (1991)	1991	Iraq	United Kingdom	Loss	0.2419	0.6121	0.0	0.0679	0.2426
Azerbaijani-Armenian War	1993	Azerbaijan	Armenia	Loss	0.4588	0.6818	1.0	0.9445	0.6638
Kargil War	1999	Pakistan	India	Loss	0.2023	0.3122	0.0357	0.0801	0.1324
Kosovo War	1999	United States of America	Yugoslavia	Win	0.9944	0.9342	0.9976	0.9931	0.9629

Table A1. Complete training dataset (151 decisive directed dyads). All ratio values rounded to 4 decimal places.